Jupyter Notebooks and Google Colab for Python/ML

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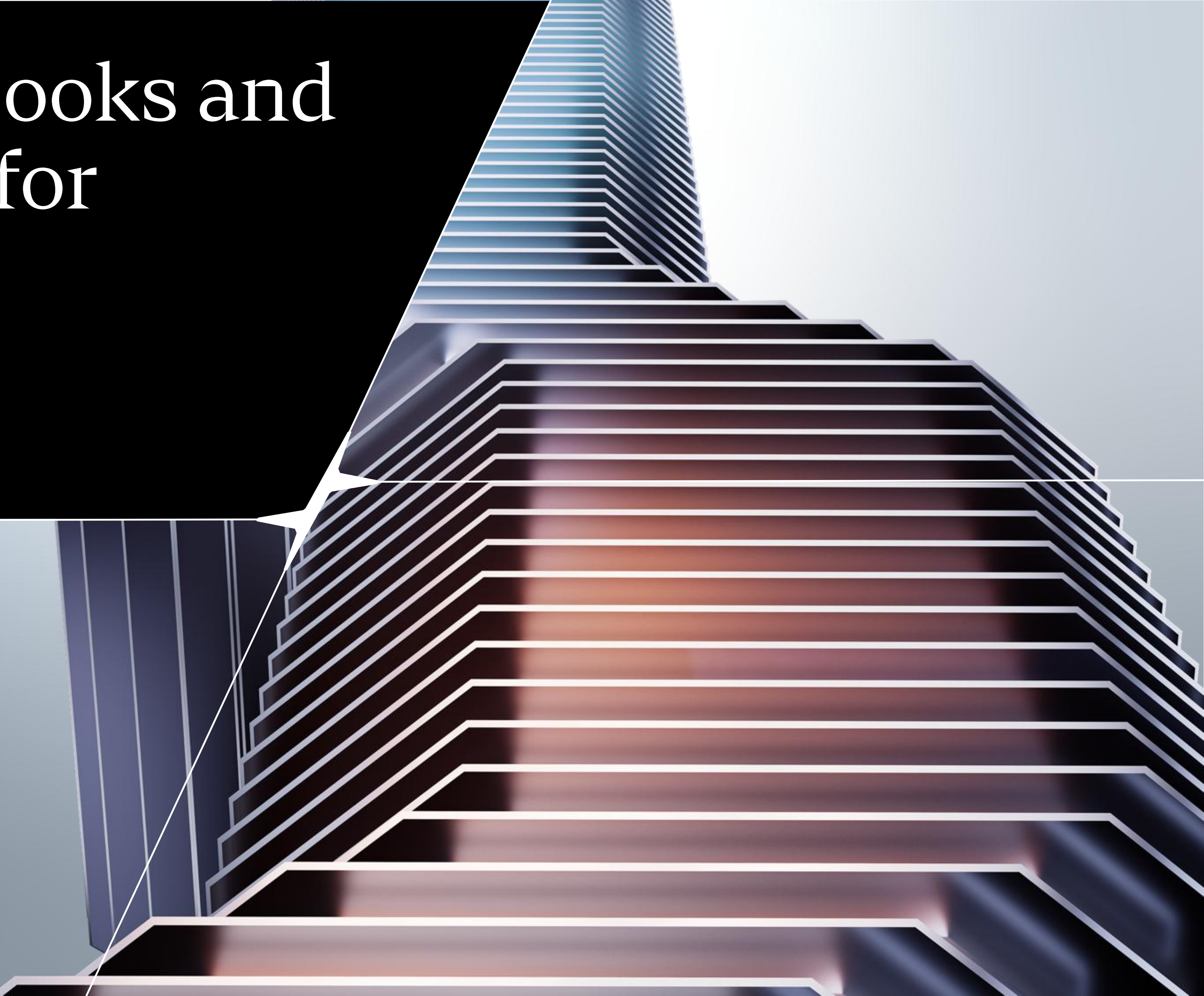
SOFTSERVE ACADEMY MENTOR



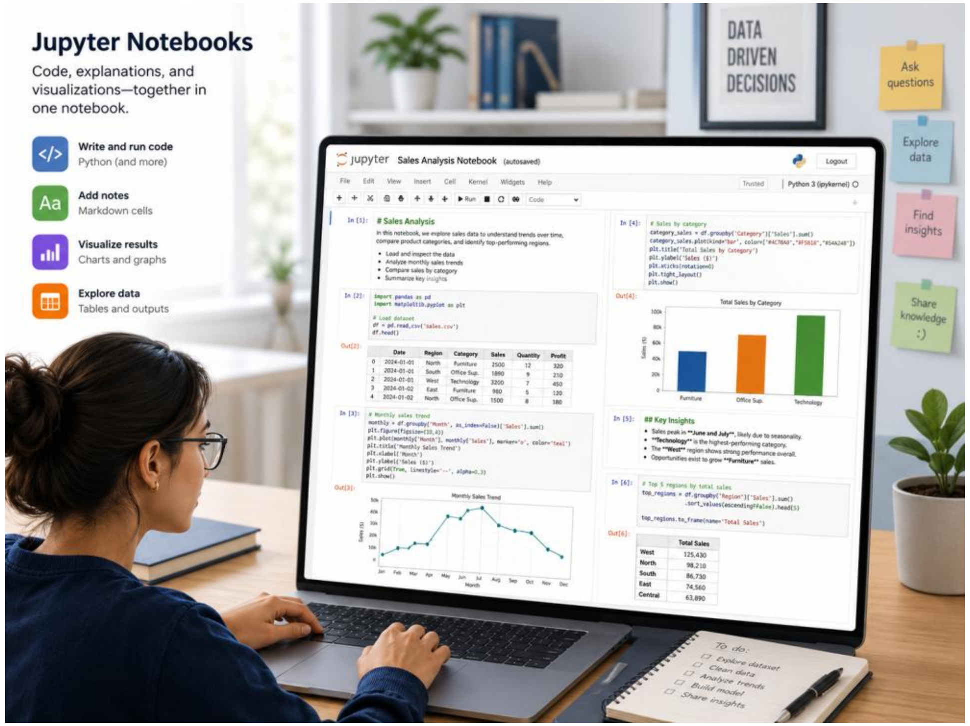
SoftServe

WHY NOTEBOOKS MATTER IN ML?

- Notebooks **combine** code, explanation, outputs, equations, charts, and experimentation in one place.

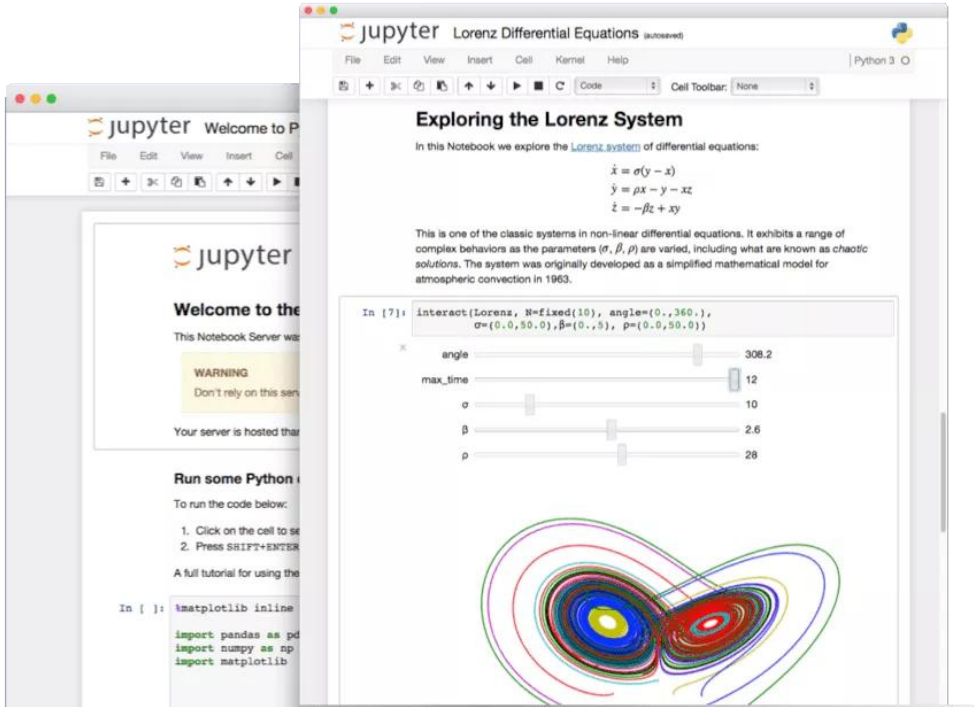
- For ML **beginners**, they are ideal for learning, exploring data, and documenting thinking.

- Notebooks are not the media to deploy models in **production** environments.



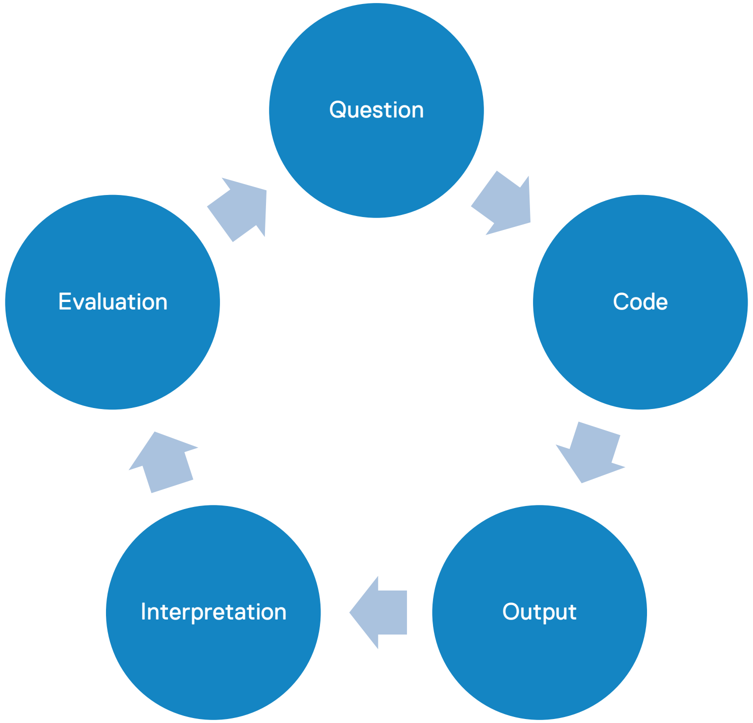
WHAT IS A JUPYTER NOTEBOOK?

- A notebook is an **interactive document** made of cells: code cells, Markdown cells, outputs, visualizations, and rich media.
- Files usually use the **.ipynb** format.
- They have become a **standard** when running Data Science and ML experiments.



WHY DO NOTEBOOKS EXIST?

- They support **interactive** computing: write a little code, run it, inspect the result, explain what happened, and continue.
- This makes them useful for **exploration** and **communication**.
- They leverage the interactive nature of the Python **interpreter**.

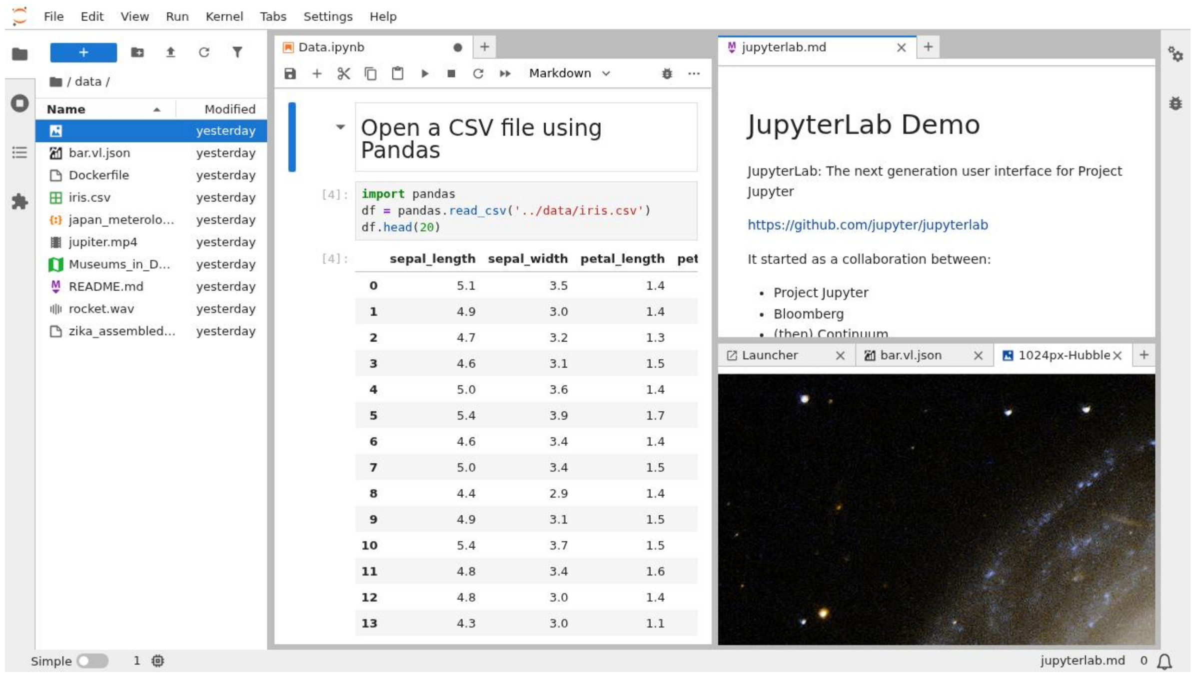


ANATOMY OF A NOTEBOOK

Key parts:

- Cells
- Kernel
- Runtime
- Outputs
- Variables in memory
- File browser
- Environment/packages.

The notebook document is not the same as the running kernel.



USING NOTEBOOKS OR NOT?

They are good for...

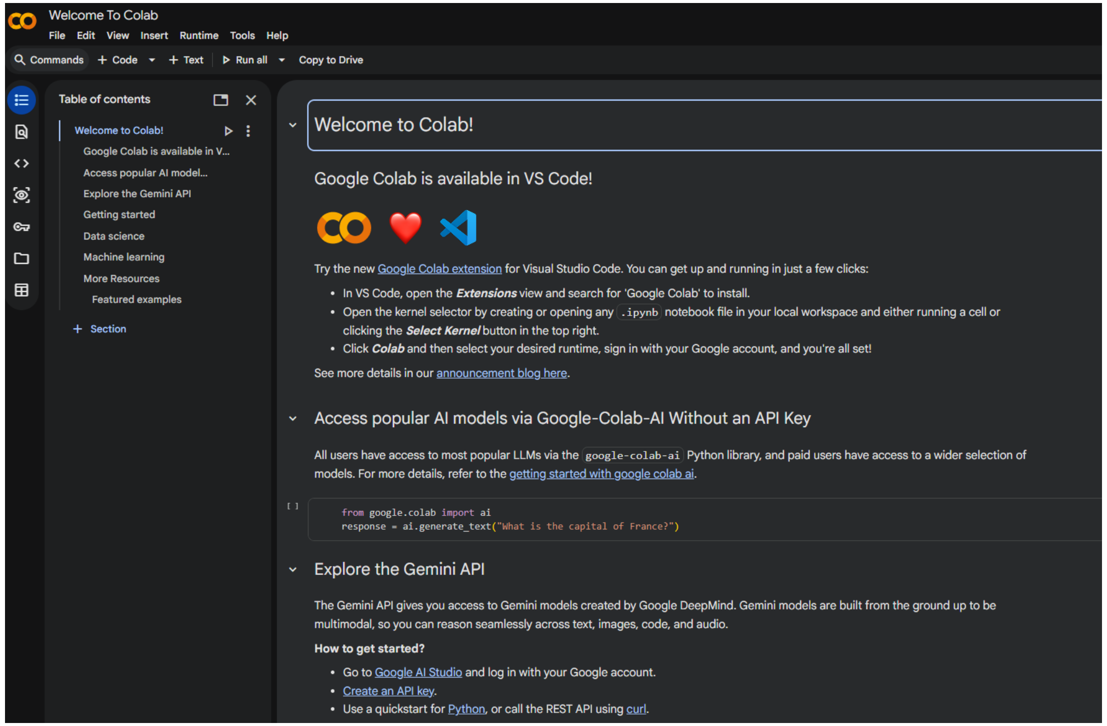
- Data exploration
- Visualizations
- Small experiments
- ML demos
- Tutorials
- Teaching
- Reports
- Reproducible walkthroughs

But, they are not that good for...

- Large production systems
- Complex software architecture
- Hidden execution order
- Hard-to-review long notebooks
- Fragile environments

GOOGLE COLAB: NOTEBOOKS IN THE CLOUD

- Colab is a **hosted** Jupyter Notebook service.
- It requires **no local setup** and provides access to **cloud compute**, including GPUs/TPUs, with usage limits.
- Requires a **Google account** and allows persistence using Google Drive.



JUPYTER VS GOOGLE COLAB

Local Jupyter

- More control
- Local files
- Custom environments
- Works offline after setup

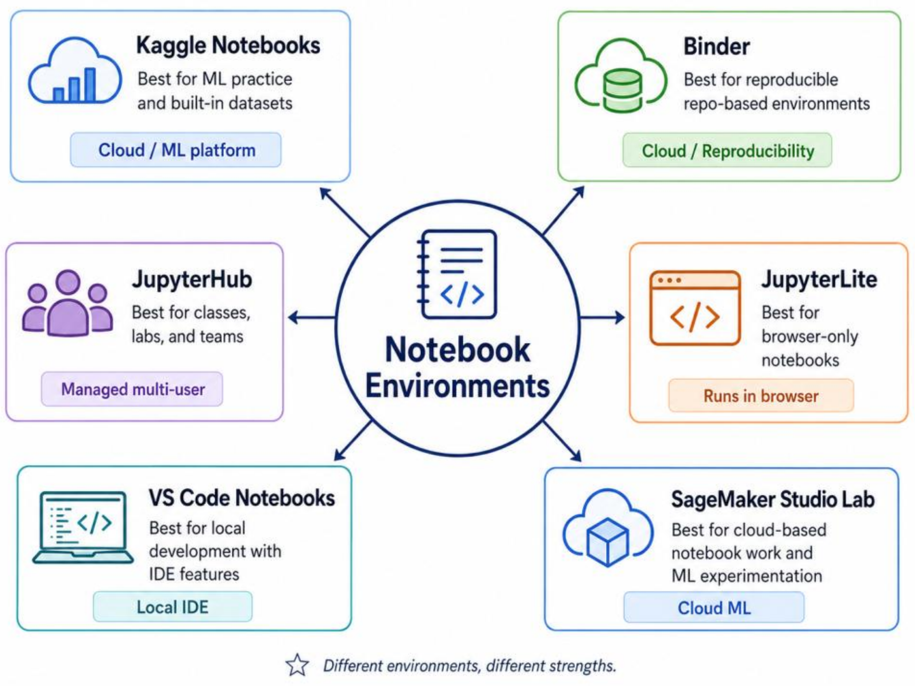
Give it a try here: https://jupyter.org/try-jupyter/notebooks

Google Colab

- Fast start
- Easy sharing
- Cloud GPUs
- Google Drive integration
- Less control and runtime/session limits

Give it a try here: https://colab.research.google.com/notebooks/welcome.ipynb

OTHER NOTEBOOK-BASED ENVIRONMENTS



SOME USEFUL RESOURCES

- Jupyter Project: https://jupyter.org/
- Jupyter Notebooks: https://docs.jupyter.org/en/latest
- Notebook Narratives: https://docs.jupyter.org/en/stable/use/use-cases/narrative-notebook.html
- Installing Jupyter: https://jupyter.org/install
- Google Colab: https://colab.research.google.com/
- Kaggle Notebooks: https://www.kaggle.com/docs/notebooks
- Jupyter Binder: https://jupyter.org/binder
- Jupyter Notebooks in VS Code: https://code.visualstudio.com/docs/datascience/jupyter-notebooks
- JupyterLite: https://jupyterlite.readthedocs.io/en/stable

Thanks!

Any further questions?

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